

Solution Architecture

Project Name : ShopEZ – MERN Stack E-Commerce Application

Phase: Project Design Phase

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Date: 17 June 2026

1. Overview

The ShopEZ E-Commerce Application is developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The architecture follows a client-server model where users interact with the React frontend, which communicates with backend APIs built using Express.js and Node.js. All application data is stored in MongoDB.

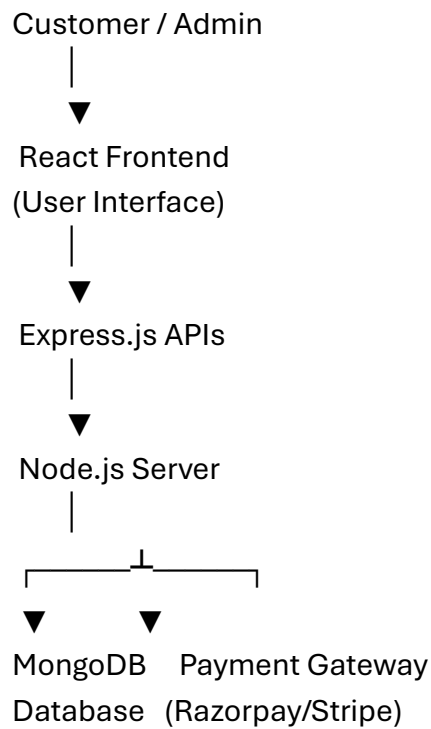
2. Objectives

- Provide a secure online shopping platform.
- Enable efficient product browsing and management.
- Support user authentication and authorization.
- Facilitate secure order placement and payment processing.
- Provide an administrative dashboard for system management.

3. Technology Stack

Layer	Technology
Frontend	React.js, HTML, CSS, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT (JSON Web Token)
Payment Integration	Razorpay / Stripe
Deployment	Render / Vercel / Netlify

4. Solution Architecture Flow



5. Module Description

User Module

- User Registration
- User Login
- Profile Management
- Order History

Product Module

- Product Listing
- Product Search
- Product Filtering
- Product Details

Cart Module

- Add to Cart
- Update Quantity
- Remove Product

Order Module

- Place Order
- Order Tracking
- Order Management

Payment Module

- Secure Online Payment
- Payment Verification
- Transaction Processing

Admin Module

- Product Management
 - User Management
 - Order Management
 - Sales Monitoring
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6. Data Flow

1. User accesses the ShopEZ website.
 2. React frontend sends requests to Express APIs.
 3. Node.js processes business logic.
 4. MongoDB stores and retrieves data.
 5. Payment Gateway processes transactions.
 6. Order details are stored and displayed to users.
 7. Admin manages products, users, and orders through the dashboard.
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7. Security Features

- JWT Authentication
- Password Encryption
- Protected Routes
- Secure Payment Gateway Integration

- Input Validation and Error Handling
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8. Advantages of the Architecture

- Scalable and flexible.
 - Fast and responsive user interface.
 - Secure authentication and payment processing.
 - Easy maintenance and future enhancements.
 - Supports increasing numbers of users and products.
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Conclusion

The MERN-based architecture of ShopEZ provides a secure, scalable, and efficient e-commerce solution. The architecture supports seamless interaction between users, administrators, backend services, and the database while ensuring performance and reliability.